higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

FLUID MECHANICS N5

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This marking guideline consists of 10 pages.

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QUESTION 1

1.1 The purpose of bearing lubrication is to prevent direct metallic (metal-to-metal)✓ contact between different rotating and sliding components in a mechanical mechanism.✓

Advantages:

- Eliminates rapid material expansion due to temperature rise
- Increases the expected life span of the material
- Reduces friction and wear
- Dissipates friction heat
- Prevents metal decay due to corrosion
- Protects against harmful elements
- Prevents power loss due to friction (2 + any 2) (4)

1.2 The absolute coefficient of viscosity of a fluid is the tendency to resist sliding layers of fluid measured in Pa.s.✓ The kinetic viscosity is the ratio of the absolute viscosity✓ to the mass density measured in m^2/s.✓ (1 + 2) (3)

1.3 1.3.1

$$\text{From } F=\frac{\mu Av}{t}$$

$$t=\frac{D-d}{2}$$

$$=\frac{0,1503-150}{2}$$

$$=150\times10^{-6}\ m \checkmark$$

$$A=\pi DL$$

$$¿\pi\times0,15\times0,17$$

$$¿80,111\times1^{-3}\ m^2 \checkmark$$

$$v=\pi DN$$

$$¿\pi\times0,15\times20 \checkmark$$

$$¿9,425\ \text{m/s} \checkmark$$

$$F=\frac{\mu Av}{t}$$

$$¿\frac{0,02\times80,111\times1^{-3}\times9,425}{150\times10^{-6}} \checkmark$$

$$¿100,67\ N \checkmark$$

$$P=Fv$$

$$¿100,67\times9,425 \checkmark$$

$$¿948,762\ W \checkmark$$

(8)

1.3.2
$$v=\frac{\mu}{\rho}$$
$$=\frac{0,02}{830}\checkmark$$
$$=24,096\times10^{-6}\ m^2/s\checkmark$$ (2)

1.3.3
$$P=Fv$$
$$t=150,03-150$$
$$=0,0003\ m\checkmark$$

$$P=\frac{0,02\times80,111\times1^{-3}\times5}{0,0003}\checkmark$$
$$\therefore 26,704\ N\checkmark$$ (3)
[20]

QUESTION 2

2.1 According to Pascal's law, an external force exerted on a unit area of a closed hydraulic system will be equally transmitted✓ with no loss in energy to every unit area of the interior of the vessel.✓ (2)

2.2 A hydraulic accumulator is a device used to store fluid energy under a head and supply it during a high demand from the system. (1)

2.3 2.3.1 consider slave cylinder
from $P_s=P_m$
$$F_s=\frac{F_m D^2}{d^2}\checkmark$$
$$\therefore\frac{800\times0,1^2}{0,035^2}\checkmark$$
$$\therefore 6,531\ \text{kN}\checkmark$$ (3)

2.3.2

$$P_m = \frac{F}{A}$$

$$¿ \frac{4 \times 800}{\pi \times 0,035^2} \checkmark$$

$$¿ 831,503 \text{ kPa} \checkmark$$

$$\text{from } \frac{1}{K_e} = \frac{1}{K_{fluid}} + \frac{1}{K_{cyl}} + \frac{V_{air}}{V_{total} K_{air}}$$

$$K_{cyl} = \frac{50 \times 10^9}{2,5}$$

$$¿ 20 \text{ GPa} \checkmark$$

$$\text{now } K_e = \frac{1}{K_{fluid}} + \frac{1}{K_{cyl}} + 0 \checkmark$$

$$¿ \frac{1}{1,8 \times 10^9} + \frac{1}{20 \times 10^9} \checkmark$$

$$\therefore K_e = 1,651 \text{ GPa} \checkmark$$

$$\varepsilon = \frac{P}{K_e}$$

$$¿ \frac{500 \times 10^{-6} \times 831,503 \times 10^3}{1,651 \times 10^9} \checkmark$$

$$\therefore \varepsilon = 503,521 \times 10^{-6} \ m^3 \checkmark$$

$$V_{total} = 3 \times \frac{\pi}{4} \times 0,1^2 \times 0,02 \checkmark$$

$$¿ 471,239 \times 10^{-6} \ m^3 \checkmark$$

$$\Delta V = \varepsilon \times V_{total}$$

$$¿ 503,521 \times 10^{-6} \times 471,239 \times 10^{-6} \checkmark$$

$$\therefore = 237,3 \times 10^{-9} \ m^3 \checkmark$$

$$\therefore \Delta L_{master} = \frac{\Delta V}{A}$$

$$¿ \frac{4 \times 237,3 \times 10^{-9}}{0,035^2} \checkmark$$

$$¿ 246,623 \times 10^{-6} \ m \checkmark$$

(14)
[20]

QUESTION 3

3.1 consider curved side

$$\bar{y} = \frac{D}{2}$$

$$= \frac{2}{2} \checkmark$$

$$= 1\ m \checkmark$$

$$A = \pi Dh \checkmark$$

$$= \pi \times 1 \times 2 \checkmark$$

$$= 6,283\ m$$

$$F = \rho g \bar{y} A$$

$$= 900 \times 9,81 \times 1 \times 6,283 \checkmark$$

$$= 55,473\ \text{kN} \checkmark$$

(6)

3.2 consider circular side

$$\bar{y} = \frac{D}{2}$$

$$= \frac{1,45}{2}$$

$$= 0,725\ m \checkmark$$

$$A = \pi Dh$$

$$= \frac{\pi}{4} \times 1,45^2 \checkmark$$

$$= 1,651\ m \checkmark$$

$$F = \rho g \bar{y} A$$

$$= 900 \times 9,81 \times 0,725 \times 1,651 \checkmark$$

$$= 6,94\ \text{kN} \checkmark$$

(5)

3.3

$$\bar{h} = \frac{2}{3} D$$

$$= \frac{2}{3} \times 1,45 \checkmark$$

$$= 0,967\ m \checkmark$$

(2)

3.4 $W_{object}+F=F_{total}$

$F_{bouyancy}=(66\times9,81)+\left(900\times9,81\times\frac{\pi}{4}\times1,45^2\times2\right)$✓✓

¿$647,46+29\ 158,649$ ✓

¿$29,806$ kN ✓

$F_{total}=F_{bouyancy}$

$29,806\times10^3=1\ 029\times9,81\times\frac{\pi}{4}\times1,45^2h$✓

$h=\frac{29,806\times10^3}{16,669\times10^3}$ ✓

¿$1,788\ m$ ✓ (7)

[20]

QUESTION 4

4.1 The criteria used includes calculation of the Reynolds number to determine the type of flow in a fluid-flowing system✓ with the aid of equation $R=\frac{\rho vd}{\mu}$.✓ The value from this formula will determine the type of flow within the pipe system – if $R=\frac{\rho vd}{\mu}\leq2\ 000$ ✓ the flow is deemed to be lamina✓ and if $R=\frac{\rho vd}{\mu}\geq2\ 500$ ✓ the flow is deemed to be turbulent.✓ (6)

4.2 According to Bernoulli's theorem, the quantity of energy of each fluid particle is the same,✓ provided that that no energy enters or leaves the system at any point, i.e. no energy losses or gains by the fluid system.✓ (2)

4.3 4.3.1 $\frac{P_1}{\rho g}+\frac{v_1^2}{2g}+Z_1=\frac{P_2}{\rho g}+\frac{v_2^2}{2g}+Z_2$ ✓

$0+\frac{20^2}{19,62}+0=0+\frac{v_2^2}{19,62}+4,5$ ✓✓

$v_2=17,655$ m/s✓ (4)

4.3.2 from $Q_1=Q_2$

$d_2=\sqrt{\frac{v_1\times d_1^2}{v_2^2}}$ ✓

$d_2=\sqrt{\frac{20\times0,035^2}{17,655}}$✓

¿$37,252\times10^{-3}\ m$✓ (3)

4.3.3

$$P=\frac{\overset{0}{m}v^2}{2}$$

$$\text{where } \overset{0}{m}=\rho Av$$

$$A=\frac{\pi}{4}\times 0,035^2$$

$$\therefore 962,1128\times 10^{-3}\ m^2 \checkmark$$

$$\overset{0}{m}=1\ 000\times\frac{\pi}{4}\times 0,035^2\times 20\checkmark$$

$$\therefore 19,24226\ \text{kg/s}\checkmark$$

$$P=\frac{19,24226\times 20^2}{2}\ \checkmark$$

$$\therefore 3,849\ \text{kW}\checkmark$$

(5)
[20]

QUESTION 5

5.1 Two energy points are used in the fluid system as references,✓ i.e. A (energy point in the fluid channel) or B (energy point at the entrance to the tube).✓ Where the velocity at energy point B will be reduced to zero (which is now the stagnation point) when the tube is filled with fluid, this results in a velocity head rise in the tube.✓ $\Delta v=\sqrt{2gh}$.✓ (4)

5.2

$$h_l=h_m\left(\frac{\rho_m}{\rho_s}-1\right)\ \checkmark$$

$$\therefore 0,038\left(\frac{13\ 600}{1\ 000}-1\right)\checkmark$$

$$\therefore 0,4788\ m\ \checkmark$$

$$v_{ave}=C_v\sqrt{2gh_l}$$

$$v_{\max}=\sqrt{19,62\times 0,4788}=3,06497\ \text{m/s}\checkmark$$

$$\therefore 0,82\times 3,06497\checkmark$$

$$\therefore v_{\max}=2,513\ \text{m/s}\checkmark$$

$$Q=v_{\max}\times A$$

$$\therefore\frac{\pi\times 0,2^2\times 2,513}{4}\ \checkmark$$

$$\therefore Q=78,957\times 10^{-3}\ m^3/\text{s}\ \checkmark$$

(8)

5.3 5.3.1

$$v_1 = \frac{Q}{A_1}$$
$$¿ \frac{4 \times 2 \times 10^{-3}}{\pi \times 0,04^2}$$
$$¿ 1,592 \text{ m/s} ✓$$

$$v_2 = \frac{Q}{A_2}$$
$$¿ \frac{4 \times 2 \times 10^{-3}}{\pi \times 0,015^2}$$
$$¿ 11,318 \text{ m/s} ✓$$ (2)

5.3.2

$$hs_1 = \left(\frac{1}{C_c} - 1\right)^2 \times \frac{v_2^2}{2g}$$
$$¿ \left(\frac{1}{0,65} - 1\right) \times \frac{11,317^2}{19,62} ✓$$
$$¿ 1,893 \ m ✓$$

$$hs_2 = \frac{kv_2^2}{2g}$$
$$¿ \frac{3 \times 11,317^2}{19,62} ✓$$
$$¿ 16,646 \ m ✓$$

$$h_{total} = 1,893 + 16,646 ✓$$
$$¿ 18,539 \ m ✓$$ (6)

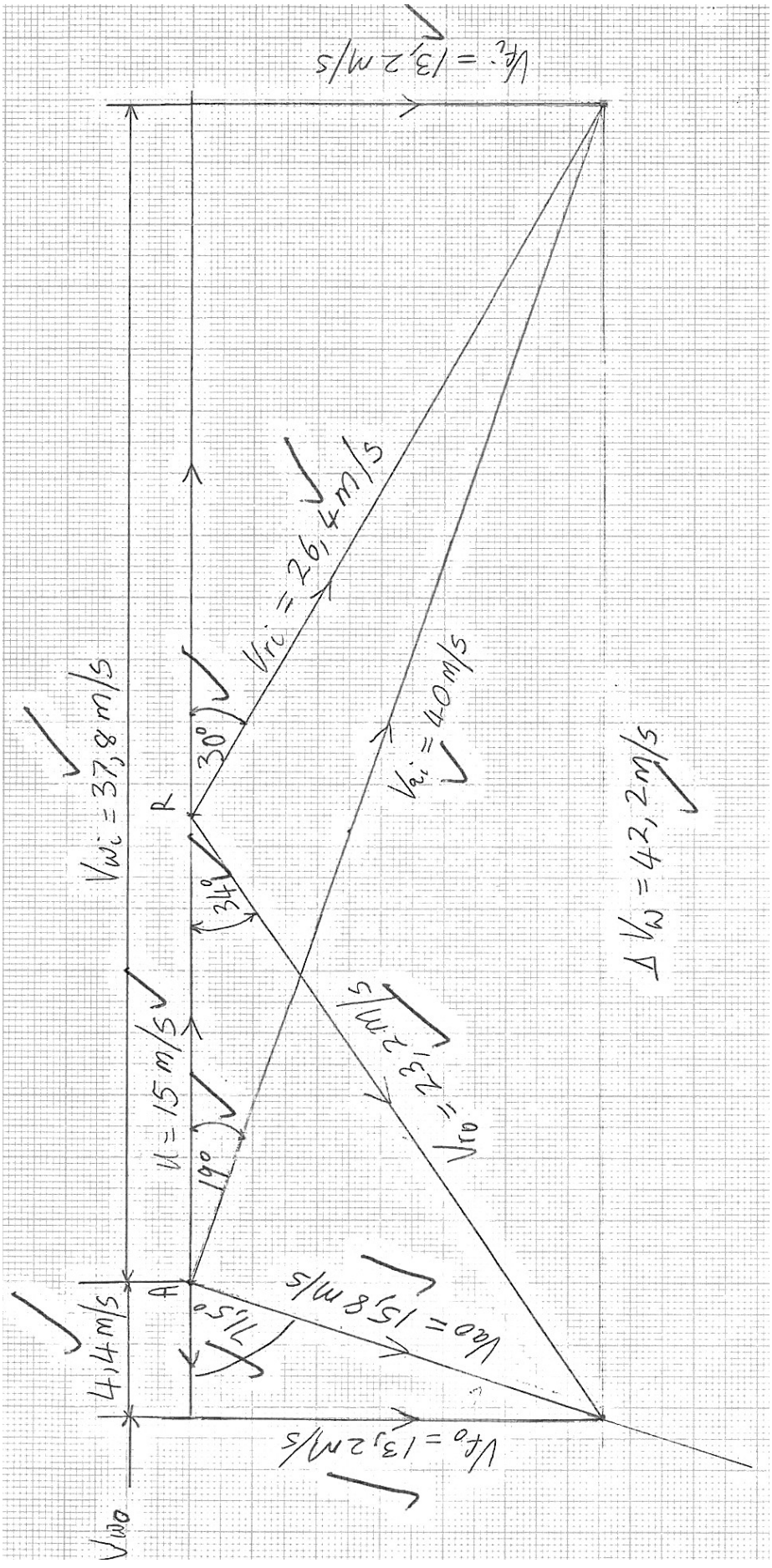
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QUESTION 6

6.1

$$U = \frac{\pi dN}{60}$$
$$= \frac{\pi \times 1,7 \times 168,517}{60}$$
$$= 15 \text{ m/s} ✓$$ (1)

6.2


$V_{fi} = 13,2\ m/s$
$V_{wi} = 37,8\ m/s$
$V_{ri} = 26,4\ m/s$
30°
R
$V_{ai} = 40\ m/s$
34°
$\Delta V_w = 42,2\ m/s$
$V_{ro} = 23,2\ m/s$
$U = 15\ m/s$
19°
A
$V_{ao} = 15,8\ m/s$
71,5°
$4,4\ m/s$
$V_{fo} = 13,2\ m/s$
V_{wo}

(14)

6.3 $P = \overset{0}{m} \Delta V_w u$

where $\Delta V_w = \Delta V_{wi} + \Delta V_{wo} = 42,2$ m/s ✓

¿ $28,5 \times 42,2 \times 15$ ✓

¿ $18,041$ kW ✓

(3)

6.4 $\eta = \dfrac{2 \Delta V_w u}{V_{ai}^2} \times 100\%$

¿ $\dfrac{2 \times 42,2 \times 15}{40^2} \times 100\%$ ✓

¿ $79,125\%$ ✓

Alternatively

$P_{in} = \dfrac{\overset{0}{m} V_{ai}^2}{2}$

¿ $\dfrac{28,5 \times 40^2}{2}$

¿ $22,8$ kW ✓

$\eta = \dfrac{18,401}{22,8} \times 100\%$

¿ $79,125\%$ ✓

(2)

[20]

TOTAL: 100

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